HIRDIANSHU SINGLA

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EDUCATION

BASc, Computer Engineering, University of British Columbia, Vancouver, BC Expected: 05/2028
Relevant Coursework: Machine Learning & Data Mining (regression, classification, clustering, neural networks, gradient descent) · Intelligent Systems (search, CSPs, planning, probabilistic reasoning) · Algorithm Design & Analysis · Relational Databases · Software Construction I & II · Operating Systems · Probability & Statistics

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02), Amazon Web Services — 2026
AWS Certified Machine Learning Engineer (MLA-C01), Amazon Web Services — In Progress

TECHNICAL PROJECTS

Terra AI — Gemini-Driven 3D Terrain Explorer *React · Three.js · TypeScript · Python · Gemini API* Oct 2026

- Built a 3D terrain explorer (React, Three.js, TypeScript, Zustand) with a Python backend that keeps the Gemini API key server-side, streams model replies and exposes function-calling tools letting the model switch layers, highlight elevation ranges and extremes, focus regions and fly the camera over real DEM data (Copernicus, Terrarium, Open-Meteo)
- Implemented the terrain analysis the model queries: Horn slope and aspect, topographic position index, D8 flow accumulation, a minimax flood-level model over a priority queue, and heuristic flood and landslide hazard indices rendered as selectable map layers
- Built the Python pipeline and web UI that fetch elevation and land-cover data for any place, classify it into terrain types, and export it as a voxel world to a custom FPGA renderer (C game on ARM Cortex-A9, Verilog GPU) over UART with compression and CRC-16 checking

Chess AI Platform *Python · PyTorch · TypeScript · Socket.io* Jan – Apr 2026

- Trained a feedforward neural network with Transformer-informed design principles on 50,000 Stockfish-evaluated positions with 70-dimensional engineered board feature vectors; tuned learning rate, dropout and batch size and analyzed convergence via R^2 and loss curves to identify underfitting and overfitting regimes
- Designed a controlled evaluation study comparing learned neural evaluation against classical minimax heuristics across 500+ test positions, quantifying accuracy against search depth and integrating both as interchangeable evaluators in a shared alpha-beta engine
- Deployed real-time two-player multiplayer via Socket.io with room-based matchmaking, server-side move validation and disconnect recovery, over a full TypeScript/Express REST API with persistent game state management

AI Financial Assistant & Messenger *Node.js · Llama3 · LLaVA · Whisper* Jan – Apr 2026

- Ranked Top 3 out of 37 groups, recognized for innovation, technical depth and real-world impact
- Built a real-time multi-room chat application (Node.js/Express/MongoDB/WebSockets) with an integrated multimodal AI panel supporting text, voice transcription (Whisper) and chart-image analysis (LLaVA) that identifies candlestick patterns, trend direction and support/resistance from uploaded screenshots
- Deployed a Llama3-powered context-aware recommendation sidebar reading live chat messages and refreshing every 30s; built a Python Flask AI server exposing six REST inference endpoints for LLM, vision and speech models with Alpha Vantage live stock price integration

Volatility Forecasting Research Pipeline *Python · GARCH · HAR-RV · Gradient Boosting* May – Aug 2026

- Built and backtested a forecasting pipeline comparing 9 statistical and ML methods (HAR-RV, GJR-GARCH, gradient-boosted trees) across 20 equities and 11 years of daily data, using Garman-Klass realized volatility (~2× more accurate than naive squared-return estimation)
- Found a leverage/market-augmented HAR model (HAR-X) significantly outperformed all baselines and neural approaches (QLIKE loss, Diebold-Mariano $p<0.01$, Holm multiple-testing correction), winning on 19/20 stocks, a result consistent with published literature
- Implemented walk-forward validation across 46 rolling windows with a tested look-ahead-bias guard preventing target and feature leakage; validated economic significance via a volatility-targeting backtest

WORK EXPERIENCE

Software Developer Intern, *Tech Exponent System, Chandigarh, Punjab* May – Aug 2026

- Developed the real-time voice controller in Python for VoiceIQ, a full-stack AI voice agent platform, handling live audio streaming and WebSocket orchestration for telephony providers, integrated with Gemini Live for streaming voice responses and tool-based conversation handling
- Collaborated with the backend team on a Node.js/Express REST API with Prisma data access, contributing to agent skill definitions (appointment booking, complaint handling, multilingual switching) dynamically loaded and activated per business through skill APIs

Trainee Analyst, *Winux Software Solutions Pvt Ltd, Mohali, Punjab* May – Aug 2025

- Designed and implemented a Python data pipeline for automated log analysis and parsing, statistical analysis of pass/fail distributions and first-fail root-cause triage report generation; applied systematic testing methodology with QA to derive edge-case scenarios improving coverage

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · C · C++ · Java · C# · Go · Rust · R · SQL · MATLAB · SystemVerilog · Verilog · VHDL · ARM assembly
Scripting, Markup & Data: Bash/Shell · PowerShell · Lua · PHP · Ruby · Tcl · GNU Make · Regex · LaTeX · HTML · CSS · JSON · YAML · XML
ML & Data: PyTorch · scikit-learn · TensorFlow · JAX · NumPy · pandas · CUDA · Neural network design & training · Backpropagation · Gradient descent · Feature engineering · Model evaluation (R^2, loss, accuracy) · Regression · Classification · Clustering · Transformer architecture · Time-series forecasting · Statistical hypothesis testing (Diebold-Mariano, Holm) · Walk-forward validation · GARCH/HAR-RV volatility modeling
AI Systems: LLM function calling & tool use (Gemini, Gemini Live, Llama3) · Streaming inference · Whisper speech-to-text · LLaVA vision models · Multimodal pipelines · Prompt design · Retrieval & context management · Model Context Protocol · Claude-assisted development
Software & Systems: Git · Linux · GDB · REST APIs · WebSockets · Node.js · Express · Flask · React · Three.js · Zustand · Vite · Prisma · MongoDB · Docker · Unit testing · CI/CD pipelines · Terraform · Infrastructure as Code · Agile/Scrum · AWS · Azure · PowerBI · ServiceNow · Microsoft 365 & Copilot
Hardware & Robotics: ROS2 · LiDAR & point cloud processing · OpenCV · Sensor fusion · Motion planning (MPCC) · Quartus · ModelSim/Questa · Verilator · Cyclone V · DE1-SoC · ARM Cortex-A9 · Arduino